

B5 General Relativity & Cosmology — Model Solutions, 2016

Question 1 — Schwarzschild orbits and circular geodesics

Problem. Schwarzschild metric. Reduce to equatorial plane; find $\dot{t}$, $\dot{\phi}$ and the orbit equation $u'' + u = A + Bu^2$. A satellite is launched azimuthally from a stationary station at $r = R$ with locally measured speed v_{circ} giving a circular orbit. Find v_{circ} and the time before the orbiting satellite collides with the station. Sketch the perturbed orbit when the launch speed is $v_{\text{circ}} + \Delta v$.

(a) Equatorial reduction and conserved quantities

The metric is spherically symmetric, so it is invariant under $\theta \rightarrow \pi - \theta$. Pick initial conditions with $\theta = \pi/2$, $\dot{\theta} = 0$ (always possible by rotation): the Euler–Lagrange equation for θ has the right-hand side proportional to $\sin \theta \cos \theta \dot{\phi}^2$, vanishing on the equator. Hence $\theta(\tau) \equiv \pi/2$ is a geodesic; restricting to it loses no generality.

Lagrangian per unit rest mass on the equator (using $\dot{} = d/d\tau$):

$$\mathcal{L} = -\left(1 - \frac{2GM}{c^2 r}\right)c^2 \dot{t}^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2. \quad (1)$$

$\mathcal{L}$ has no explicit t or ϕ :

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \dot{t}} &= -2c^2 \left(1 - \frac{2GM}{c^2 r}\right) \dot{t} = \text{const}, \\ \frac{\partial \mathcal{L}}{\partial \dot{\phi}} &= 2r^2 \dot{\phi} = \text{const}. \end{aligned}$$

Define the two constants of motion (energy E and angular momentum ℓ per unit rest mass):

$$\boxed{\left(1 - \frac{2GM}{c^2 r}\right) \dot{t} = \frac{E}{c^2}, \quad r^2 \dot{\phi} = \ell.} \quad (2)$$

(b) Radial equation and orbit equation

Normalisation of the 4-velocity, $g_{ab} \dot{x}^a \dot{x}^b = -c^2$, on the equator:

$$-\left(1 - \frac{2GM}{c^2 r}\right)c^2 \dot{t}^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2 = -c^2.$$

Substitute (??):

$$-\frac{E^2/c^2}{1 - 2GM/(c^2 r)} + \frac{\dot{r}^2}{1 - 2GM/(c^2 r)} + \frac{\ell^2}{r^2} = -c^2.$$

Multiply by $1 - 2GM/(c^2 r)$ and rearrange:

$$\boxed{\dot{r}^2 = \frac{E^2}{c^2} - \left(1 - \frac{2GM}{c^2 r}\right) \left(c^2 + \frac{\ell^2}{r^2}\right).} \quad (3)$$

Change variable to $u(\phi) = 1/r$. Use $\dot{r} = (dr/d\phi)\dot{\phi} = -u^{-2}u' \ell u^2 = -\ell u'$:

$$\dot{r}^2 = \ell^2(u')^2.$$

Insert into (??):

$$\ell^2(u')^2 = \frac{E^2}{c^2} - c^2 + \frac{2GM}{c^2}c^2u - \ell^2u^2 + \frac{2GM}{c^2}\ell^2u^3.$$

Differentiate w.r.t. ϕ and divide by $2u'$:

$$\ell^2u'' = GM - \ell^2u + 3GMu^2.$$

Divide by ℓ^2 :

$$u'' + u = A + Bu^2, \quad A = \frac{GM}{\ell^2}, \quad B = \frac{3GM}{c^2}. \quad (4)$$

(c) Circular orbit launched from the station

Circular orbit: $u = u_0 = 1/R$ constant, so (??) gives $u_0 = A + Bu_0^2$:

$$\frac{1}{R} = \frac{GM}{\ell^2} + \frac{3GM}{c^2 R^2}.$$

Solve for ℓ^2 :

$$\ell^2 = \frac{GM R^2}{R - 3GM/c^2} = \frac{GMR^2}{R(1 - 3GM/(c^2 R))}.$$

Cleaner form:

$$\ell^2 = \frac{GMR}{1 - 3GM/(c^2 R)}. \quad (5)$$

The astronaut's locally measured speed is the proper distance moved per unit proper time. The station is static at $r = R$, so the astronaut's frame uses

$$d\ell_{\perp} = R d\phi, \quad d\tau_{\text{stn}} = \sqrt{1 - \frac{2GM}{c^2 R}} dt.$$

The satellite passes the station with $\dot{r} = 0$, $\dot{\theta} = 0$, $\dot{\phi} \neq 0$. Local speed:

$$v_{\text{circ}} = \frac{R d\phi}{d\tau_{\text{stn}}} = \frac{R \dot{\phi}}{\sqrt{1 - 2GM/(c^2 R)} \dot{t}}.$$

From (??), $\dot{\phi}/\dot{t} = \ell c^2/(R^2 E) \cdot (1 - 2GM/(c^2 R))$. Use the alternative route: write $v_{\text{circ}}^2 = (R d\phi/d\tau_{\text{stn}})^2$. We have

$$\frac{(R d\phi)^2}{d\tau_{\text{stn}}^2} = \frac{R^2 \dot{\phi}^2}{(1 - 2GM/(c^2 R)) \dot{t}^2}.$$

Multiply top and bottom by $d\tau^2$ and use $r^2 \dot{\phi} = \ell$:

$$v_{\text{circ}}^2 = \frac{\ell^2/R^2}{(1 - 2GM/(c^2 R)) \dot{t}^2/d\tau^2} \cdot \frac{1}{R^2/R^2}.$$

Cleaner: along the circular orbit, the 4-velocity normalisation reads

$$-c^2 = -\left(1 - \frac{2GM}{c^2 R}\right)c^2 \dot{t}^2 + R^2 \dot{\phi}^2.$$

Solve for $R^2\dot{\phi}^2$:

$$R^2\dot{\phi}^2 = \left(1 - \frac{2GM}{c^2 R}\right)c^2\dot{t}^2 - c^2.$$

Divide by $(1 - 2GM/(c^2 R))\dot{t}^2$:

$$v_{\text{circ}}^2 = \frac{R^2\dot{\phi}^2}{(1 - 2GM/(c^2 R))\dot{t}^2} = c^2 - \frac{c^2}{(1 - 2GM/(c^2 R))\dot{t}^2}.$$

Use (??): $(1 - 2GM/(c^2 R))\dot{t} = E/c^2$, so $(1 - 2GM/(c^2 R))\dot{t}^2 = E^2/[c^4(1 - 2GM/(c^2 R))]$. Cleaner: rewrite using $\ell = R^2\dot{\phi}$:

$$v_{\text{circ}}^2 = \frac{\ell^2/R^2}{(1 - 2GM/(c^2 R))\dot{t}^2}.$$

Expand $\dot{t}^2$ from the same normalisation: $\dot{t}^2 = (1 + \ell^2/(c^2 R^2))/(1 - 2GM/(c^2 R))$. Insert:

$$v_{\text{circ}}^2 = \frac{\ell^2/R^2}{(1 - 2GM/(c^2 R)) \cdot (1 + \ell^2/(c^2 R^2))/(1 - 2GM/(c^2 R))} = \frac{\ell^2/R^2}{1 + \ell^2/(c^2 R^2)}.$$

Substitute (??):

$$\frac{\ell^2}{R^2} = \frac{GM/R}{1 - 3GM/(c^2 R)}.$$

Numerator divided by $1 + \ell^2/(c^2 R^2)$. Compute $\ell^2/(c^2 R^2) = (GM/(c^2 R))/(1 - 3GM/(c^2 R))$. Hence

$$1 + \frac{\ell^2}{c^2 R^2} = \frac{1 - 3GM/(c^2 R) + GM/(c^2 R)}{1 - 3GM/(c^2 R)} = \frac{1 - 2GM/(c^2 R)}{1 - 3GM/(c^2 R)}.$$

Divide:

$$v_{\text{circ}}^2 = \frac{GM/R}{1 - 3GM/(c^2 R)} \cdot \frac{1 - 3GM/(c^2 R)}{1 - 2GM/(c^2 R)}.$$

Simplify:

$$\boxed{v_{\text{circ}}^2 = \frac{GM/R}{1 - 2GM/(c^2 R)}}. \quad (6)$$

Note the Newtonian limit $v_{\text{circ}}^2 \rightarrow GM/R$ when $2GM/(c^2 R) \ll 1$.

Time to collision (in the astronaut's proper time). Orbital period in coordinate time: $T_t = 2\pi/\dot{\phi} \cdot (\dot{t}/1)$, but easier to relate proper times directly. Proper-time period of the satellite over one revolution:

$$T_{\text{sat}} = 2\pi/\dot{\phi} = 2\pi R^2/\ell.$$

Substitute (??):

$$T_{\text{sat}} = 2\pi R^2/\sqrt{GMR/(1 - 3GM/(c^2 R))} = 2\pi\sqrt{R^3/GM} \sqrt{1 - 3GM/(c^2 R)}.$$

Convert to station proper time via $d\tau_{\text{stn}} = \sqrt{1 - 2GM/(c^2 R)} dt$ and $dt/d\tau_{\text{sat}} = \dot{t}$. Use (??):

$$\dot{t} = \frac{E/c^2}{1 - 2GM/(c^2 R)}.$$

Energy from $\dot{t}^2 = (1 + \ell^2/(c^2 R^2))/(1 - 2GM/(c^2 R))$ and $E = c^2(1 - 2GM/(c^2 R))\dot{t}$:

$$E^2/c^4 = (1 - 2GM/(c^2 R))(1 + \ell^2/(c^2 R^2)) = \frac{(1 - 2GM/(c^2 R))^2}{1 - 3GM/(c^2 R)}.$$

Coordinate-time period: $T_t = T_{\text{sat}} \dot{t} = T_{\text{sat}} \cdot E/[c^2(1 - 2GM/(c^2R))]$. Combine:

$$T_t = 2\pi\sqrt{R^3/GM} \sqrt{1 - 3GM/(c^2R)} \cdot \frac{E/c^2}{1 - 2GM/(c^2R)}.$$

Insert $E/c^2 = (1 - 2GM/(c^2R))/\sqrt{1 - 3GM/(c^2R)}$:

$$T_t = 2\pi\sqrt{R^3/GM}.$$

This is exactly Kepler's third law in coordinate time. Now convert to station proper time:

$$T_{\text{stn}} = T_t \sqrt{1 - 2GM/(c^2R)}.$$

$$\boxed{T_{\text{stn}} = 2\pi\sqrt{\frac{R^3}{GM}} \sqrt{1 - \frac{2GM}{c^2R}}.} \quad (7)$$

Newtonian limit: $T_{\text{stn}} \rightarrow 2\pi\sqrt{R^3/GM}$. Strong-field correction: time runs slower at the station than at infinity, so the wait is shorter than Kepler's value.

(d) Perturbed launch

For $|\Delta v| \ll v_{\text{circ}}$ the orbit is a near-circular ellipse: writing $u = u_0 + \delta u$ in (??) and linearising,

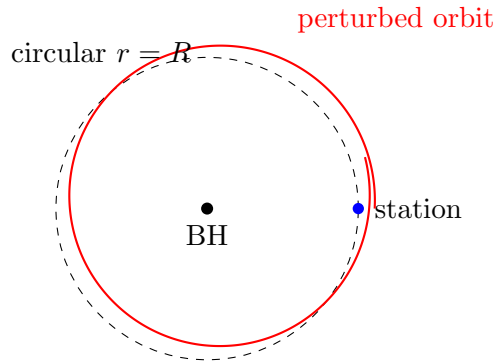
$$\delta u'' + (1 - 2Bu_0) \delta u = 0.$$

Substitute $Bu_0 = 3GM/(c^2R)$:

$$\delta u'' + (1 - \frac{6GM}{c^2R}) \delta u = 0.$$

Apsidal precession frequency (per orbit) is $\sqrt{1 - 6GM/(c^2R)} < 1$, so radial oscillations close in azimuthal angle $2\pi/\sqrt{1 - 6GM/(c^2R)} > 2\pi$: each radial period the perihelion advances by

$$\Delta\phi_{\text{prec}} \approx \pi \frac{6GM}{c^2R}.$$



The orbit oscillates between $r_- < R < r_+$, with the line of apsides precessing forward; for $\Delta v \neq 0$ the satellite returns to $r = R$ but at a shifted azimuth, so it does *not* hit the station on the first pass. However the perihelion drifts secularly, and after $\sim 2\pi/\Delta\phi_{\text{prec}}$ radial periods the apse returns to the station's azimuth: the station is therefore *not* safe forever — collision occurs after a finite (but long) time. (Strictly true outside the ISCO at $r = 6GM/c^2$; below this the perturbation is unstable and collision is rapid.)

Question 2 — Einstein equations and TOV; uniform-density star

Problem. Argue $G_{ab} = R_{ab} - \frac{1}{2}g_{ab}R$. For a static spherical metric reduce $G_{ab} = -(8\pi G/c^4)T_{ab}$ to the TOV equations. For a uniform-density star derive $p_0(R)$ and the Buchdahl bound $2GM/(c^2R) < 8/9$. Comment on a redshift $z = 3.1$ neutron star.

(a) Form of the Einstein tensor

Energy-momentum conservation in curved space requires $\nabla_a T^{ab} = 0$. If $G^{ab} \propto T^{ab}$ then $\nabla_a G^{ab} = 0$.

The Ricci tensor alone fails: the contracted Bianchi identity gives $\nabla_a R^{ab} = \frac{1}{2}\nabla^b R \neq 0$. Subtract the trace to enforce conservation:

$$G_{ab} = R_{ab} - \frac{1}{2}g_{ab}R.$$

Bianchi identity then yields $\nabla^a G_{ab} \equiv 0$. G_{ab} is symmetric (both R_{ab} and g_{ab} are) and built from the metric and its first two derivatives, matching the second-derivative structure of T_{ab} (Newtonian limit: Poisson). The proportionality constant is fixed by Newtonian limit to $-8\pi G/c^4$:

$$G_{ab} = R_{ab} - \frac{1}{2}g_{ab}R = -\frac{8\pi G}{c^4}T_{ab}.$$

(b) Static fluid: 4-velocity components

In the given coordinates G_{ab} is diagonal; for a perfect fluid (??) below

$$T^{ab} = (\rho + p/c^2)u^a u^b + g^{ab}p, \quad (1)$$

gives off-diagonal $T^{0i} \propto u^0 u^i$. Diagonality of G_{ab} forces $u^1 = u^2 = u^3 = 0$. The fluid is comoving with the static frame.

Normalisation $u^a u_a = -c^2$:

$$g_{00}(u^0)^2 = -c^2 \Rightarrow -e^{2\Phi}(u^0)^2 = -c^2.$$

Solve:

$$u^0 = c e^{-\Phi}, \quad u_0 = g_{00}u^0 = -c e^{\Phi}.$$

(c) The 00 equation: enclosed-mass form of $\Lambda(r)$

$T_{00} = (\rho + p/c^2)u_0 u_0 + g_{00}p = (\rho c^2) e^{2\Phi}$. (Cross term: $(\rho + p/c^2)c^2 e^{2\Phi} - p e^{2\Phi} = \rho c^2 e^{2\Phi}$.) Field equation:

$$G_{00} = -\frac{8\pi G}{c^4}\rho c^2 e^{2\Phi}.$$

Substitute the given G_{00} :

$$-\frac{e^{2\Phi}}{r^2} \frac{d}{dr} [r(1 - e^{-2\Lambda})] = -\frac{8\pi G}{c^2} \rho e^{2\Phi}.$$

Cancel $e^{2\Phi}$ and rearrange:

$$\frac{d}{dr} [r(1 - e^{-2\Lambda})] = \frac{8\pi G}{c^2} \rho r^2.$$

Integrate from 0 to r with $r(1 - e^{-2\Lambda}) \rightarrow 0$ at the origin:

$$r(1 - e^{-2\Lambda}) = \frac{8\pi G}{c^2} \int_0^r \rho(r') r'^2 dr' = \frac{2GM(r)}{c^2}.$$

Solve for $e^{-2\Lambda}$:

$$e^{-2\Lambda(r)} = 1 - \frac{2GM(r)}{c^2 r}. \quad (8)$$

(d) The 11 equation: $d\Phi/dr$

$T_{11} = g_{11}p = e^{2\Lambda}p$. Field equation:

$$G_{11} = -\frac{8\pi G}{c^4}e^{2\Lambda}p.$$

Substitute given G_{11} :

$$\frac{e^{2\Lambda}}{r^2} - \frac{1}{r^2} - \frac{2\Phi'}{r} = -\frac{8\pi G}{c^4}e^{2\Lambda}p.$$

Multiply by $r^2e^{-2\Lambda}$:

$$1 - e^{-2\Lambda} - 2r\Phi'e^{-2\Lambda} = -\frac{8\pi G}{c^4}r^2p \cdot (\text{wait: re-multiply carefully}).$$

Restart: rearrange the original equation algebraically. Multiply by r :

$$\frac{e^{2\Lambda}}{r} - \frac{1}{r} - 2\Phi' = -\frac{8\pi G}{c^4}r e^{2\Lambda}p.$$

Now multiply by $e^{-2\Lambda}$:

$$\frac{1}{r} - \frac{e^{-2\Lambda}}{r} - 2\Phi'e^{-2\Lambda} = -\frac{8\pi G}{c^4}r p.$$

Use (??): $1 - e^{-2\Lambda} = 2GM/(c^2r)$. Hence $1/r - e^{-2\Lambda}/r = 2GM/(c^2r^2)$:

$$\frac{2GM(r)}{c^2r^2} - 2\Phi'e^{-2\Lambda} = -\frac{8\pi G}{c^4}r p.$$

Solve for Φ' :

$$2\Phi'e^{-2\Lambda} = \frac{2GM(r)}{c^2r^2} + \frac{8\pi Gpr}{c^4}.$$

Divide by $2e^{-2\Lambda} = 2(1 - 2GM/(c^2r))$:

$$\boxed{\frac{d\Phi}{dr} = \frac{GM(r) + 4\pi Gpr^3/c^2}{c^2r^2(1 - 2GM(r)/(c^2r))}}. \quad (9)$$

(e) TOV pressure equation

Energy-momentum conservation (given):

$$(\rho c^2 + p) \frac{d\Phi}{dr} = -\frac{dp}{dr}.$$

Substitute (??):

$$\boxed{\frac{dp}{dr} = -(\rho c^2 + p) \frac{GM(r) + 4\pi Gpr^3/c^2}{c^2r^2(1 - 2GM(r)/(c^2r))}}. \quad (10)$$

(f) Uniform-density star: central pressure

$\rho = \text{const}$, so $M(r) = \frac{4}{3}\pi\rho r^3$ for $r \leq R$.

Convenient combination: $4\pi G\rho/c^2 = 3GM(R)/(c^2R^3) \cdot 1$ (consistent). Define

$$A \equiv \rho c^2, \quad u(r) \equiv 2GM(r)/(c^2r) = \frac{8\pi G\rho r^2}{3c^2}.$$

Plug into (??): numerator inside the bracket is $GM(r) + 4\pi G\rho r^3/c^2 = G \cdot \frac{4}{3}\pi\rho r^3 + 4\pi G\rho r^3/c^2 = \frac{4\pi G r^3}{3c^2}(\rho c^2 + 3p)$.

$$\frac{dp}{dr} = -(\rho c^2 + p) \cdot \frac{4\pi G r^3(\rho c^2 + 3p)/(3c^2)}{c^2 r^2(1-u)} = -\frac{4\pi G r}{3c^4} \cdot \frac{(\rho c^2 + p)(\rho c^2 + 3p)}{1-u}.$$

Use $1-u = 1 - 8\pi G\rho r^2/(3c^2)$. Separate variables:

$$\frac{dp}{(\rho c^2 + p)(\rho c^2 + 3p)} = -\frac{4\pi G r/(3c^4)}{1 - 8\pi G\rho r^2/(3c^2)} dr.$$

RHS is exact: let $w = 1 - 8\pi G\rho r^2/(3c^2)$, $dw = -16\pi G\rho r/(3c^2) dr$:

$$\text{RHS} = -\frac{4\pi G r}{3c^4} \cdot \frac{1}{w} dr = \frac{1}{4\rho c^2} \cdot \frac{dw}{w}.$$

Apply hint with $A = \rho c^2$, $x = p$:

$$\int \frac{dp}{(A+3p)(A+p)} = \frac{3}{2A} \int \frac{dp}{A+3p} - \frac{1}{2A} \int \frac{dp}{A+p} = \frac{1}{2A} [\ln(A+3p) - \ln(A+p)].$$

Equate with LHS integrated:

$$\frac{1}{2A} \ln \frac{A+3p}{A+p} = \frac{1}{4A} \ln w + \text{const.}$$

Multiply by $2A$:

$$\ln \frac{A+3p}{A+p} = \frac{1}{2} \ln w + \text{const.}$$

Exponentiate:

$$\frac{A+3p}{A+p} = K\sqrt{w}.$$

Boundary at the surface $r = R$: $p(R) = 0$, $w(R) = 1 - 2GM/(c^2 R)$. Hence $K = 1/\sqrt{1 - 2GM/(c^2 R)}$. Total mass $M = \frac{4}{3}\pi\rho R^3$.

At the centre $r = 0$: $w(0) = 1$, so

$$\frac{A+3p_0}{A+p_0} = \frac{1}{\sqrt{1 - 2GM/(c^2 R)}}.$$

Solve for p_0 . Cross-multiply:

$$(A+3p_0)\sqrt{1 - 2GM/(c^2 R)} = A+p_0.$$

Let $\sigma \equiv \sqrt{1 - 2GM/(c^2 R)}$:

$$A\sigma + 3p_0\sigma = A+p_0.$$

Group p_0 on one side:

$$p_0(3\sigma - 1) = A(1 - \sigma).$$

Solve and substitute $A = \rho c^2$:

$$\boxed{p_0 = \rho c^2 \frac{1 - \sqrt{1 - 2GM/(c^2 R)}}{3\sqrt{1 - 2GM/(c^2 R)} - 1}.} \quad (11)$$

(g) Buchdahl bound

$p_0 \rightarrow \infty$ when the denominator of (??) vanishes:

$$3\sqrt{1 - 2GM/(c^2 R)} = 1.$$

Square:

$$9(1 - 2GM/(c^2 R)) = 1.$$

Solve:

$$\frac{2GM}{c^2 R} = \frac{8}{9}.$$

Hence the upper bound on the mass at fixed R :

$$\boxed{M < \frac{4c^2 R}{9G}}. \quad (12)$$

(h) Plausibility of $z = 3.1$

Surface gravitational redshift in the Schwarzschild exterior:

$$1 + z = \frac{1}{\sqrt{1 - 2GM/(c^2 R)}}.$$

Solve for the compactness:

$$\frac{2GM}{c^2 R} = 1 - \frac{1}{(1 + z)^2}.$$

At $z = 3.1$, $(1 + z)^2 = 4.10^2 = 16.81$, so $1/(1 + z)^2 = 0.0595$:

$$\frac{2GM}{c^2 R} = 0.940.$$

Compare with Buchdahl (??): $8/9 = 0.889$. Since $0.940 > 0.889$, the claimed redshift exceeds the maximum allowed by any static fluid configuration with $\rho \geq 0$ and ρ non-increasing outwards.

$$\boxed{\text{Claim implausible: } 2GM/(c^2 R) = 0.94 > 8/9.}$$

Question 3 — FRW geometry and surface-brightness dimming

Problem. Realise the 3-sphere line element; state flat and hyperbolic counterparts; quote $4\pi r^2$ -style surface areas. Justify the FRW form. Derive cosmological redshift. Compute angular diameter $\Delta\theta$ and the surface-brightness scaling, and discuss telescope diameter for high- z galaxies.

(a) The 3-sphere line element

Parametrise $w^2 + x^2 + y^2 + z^2 = A^2$ via

$$w = A \cos \chi, \quad x = A \sin \chi \sin \theta \cos \phi, \quad y = A \sin \chi \sin \theta \sin \phi, \quad z = A \sin \chi \cos \theta,$$

$\chi \in [0, \pi]$.

$$dw^2 + dx^2 + dy^2 + dz^2 = A^2 [d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2)].$$

Change to $\bar{r} \equiv A \sin \chi$:

$$d\bar{r} = A \cos \chi d\chi, \quad A^2 d\chi^2 = \frac{d\bar{r}^2}{\cos^2 \chi} = \frac{A^2 d\bar{r}^2}{A^2 - \bar{r}^2}.$$

Hence

$$ds_{(+1)}^2 = \frac{A^2}{A^2 - \bar{r}^2} d\bar{r}^2 + \bar{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2).$$

Flat ($k = 0$):

$$ds_{(0)}^2 = d\bar{r}^2 + \bar{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2).$$

Hyperbolic ($k = -1$): replace $\sin \chi \rightarrow \sinh \chi$; equivalently

$$ds_{(-1)}^2 = \frac{A^2}{A^2 + \bar{r}^2} d\bar{r}^2 + \bar{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2).$$

Surface area of $\bar{r} = \bar{R}$. The induced (θ, ϕ) metric is $\bar{R}^2 (d\theta^2 + \sin^2 \theta d\phi^2)$ in all three cases:

$$\text{Area} = 4\pi \bar{R}^2 \quad (\text{flat, closed, open}).$$

Geometry differs in the radial part, not in the area element.

(b) FRW from homogeneity and isotropy

Cosmological principle: the spatial 3-geometry is maximally symmetric (isotropic at every point). The only such 3-geometries are flat ($k = 0$), 3-sphere ($k = +1$) and 3-hyperboloid ($k = -1$). Allow the overall length scale to depend on time, $a(t)$: the spatial part of the line element is $a^2(t) d\sigma_k^2$. Combining all three k cases by absorbing A into a and choosing r such that the angular part is $a^2 r^2$:

$$ds^2 = -c^2 dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right],$$

with $k = 0, \pm 1$. The cosmic time t is the proper time of a comoving observer (lapse $-c^2$). Homogeneity guarantees a depends only on t ; isotropy fixes the angular block as the round 2-sphere.

(c) Cosmological redshift

Radial null geodesics: $ds^2 = 0$ on the equator with $d\theta = d\phi = 0$:

$$c dt = \pm \frac{a(t) dr}{\sqrt{1 - kr^2}}.$$

Integrate for one wavefront $tE \rightarrow t_0$:

$$\int_{tE}^{t_0} \frac{c dt}{a(t)} = \int_0^R \frac{dr}{\sqrt{1 - kr^2}}.$$

The RHS is fixed by R (comoving). For a wavefront emitted one period later, $tE + \delta t_E \rightarrow t_0 + \delta t_0$, the comoving integral on the right is identical:

$$\frac{\delta t_E}{a(t_E)} = \frac{\delta t_0}{a(t_0)}.$$

Identify $\delta t = 1/\nu$:

$$\frac{\lambda_0}{\lambda_E} = \frac{a(t_0)}{a(t_E)}.$$

Definition $1 + z = \lambda_0/\lambda_E$:

$$\boxed{1 + z = \frac{a(t_0)}{a(t_E)}}. \quad (13)$$

(d) Angular diameter and surface brightness

Angular size. At emission time t_E , the proper transverse size of the disc is its diameter d . The proper transverse separation orthogonal to a radial null geodesic at $r = R$, $t = t_E$ is, from the FRW metric,

$$d\ell_\perp = a(t_E) R d\theta.$$

Set $d\ell_\perp = d$:

$$\boxed{\Delta\theta = \frac{d}{a(t_E)R} = \frac{d(1+z)}{a(t_0)R}}. \quad (14)$$

Define the angular-diameter distance $d_A \equiv a(t_E)R$, so $\Delta\theta = d/d_A$. (At large z , $\Delta\theta$ can *increase* with z .)

Flux and surface brightness. Two factors of $(1+z)$ enter the flux: each photon redshifts in energy by $(1+z)$, and arrival rate at the detector is reduced by another $(1+z)$ (time dilation). The proper area of the wavefront sphere at present is $4\pi a^2(t_0)R^2$.

$$F = \frac{L}{4\pi a^2(t_0)R^2(1+z)^2}.$$

The luminosity distance is $d_L = a(t_0)R(1+z)$, with $F = L/(4\pi d_L^2)$.

Solid angle subtended:

$$\Delta\Omega = \pi(\Delta\theta/2)^2 = \frac{\pi d^2}{4 a^2(t_E)R^2}.$$

Surface brightness:

$$\Sigma = \frac{F}{\Delta\Omega} = \frac{L}{4\pi a^2(t_0)R^2(1+z)^2} \cdot \frac{4 a^2(t_E)R^2}{\pi d^2}.$$

Use $a(t_E) = a(t_0)/(1+z)$:

$$\Sigma = \frac{L}{\pi^2 d^2} \cdot \frac{1}{(1+z)^4}.$$

$$\boxed{\Sigma \propto \frac{1}{(1+z)^4} \text{ (Tolman dimming)}}. \quad (15)$$

Implication for telescope diameter. A telescope of aperture D collects flux $\propto D^2$ and resolves a beam of solid angle $\Delta\Omega_{\text{beam}} \sim (\lambda/D)^2$. The signal from a resolved galaxy is the surface brightness times the beam area times D^2 :

$$S \propto \Sigma \cdot (\lambda/D)^2 \cdot D^2 = \Sigma \lambda^2.$$

The telescope-aperture factors cancel. Therefore for a galaxy that is *resolved* (extended source) increasing D does *not* improve the per-pixel brightness, and the $(1+z)^{-4}$ Tolman dimming cannot be defeated by aperture. (Larger D does help unresolved sources or pushes the resolution limit so that previously-unresolved high- z galaxies become resolved; once resolved, the limit is set by detector noise versus the dimmed surface brightness, and longer integration time, not larger aperture, is required.)

Question 4 — Thermal history: nucleosynthesis, decoupling, matter era

Problem. Compare nucleosynthesis and decoupling in the thermal history. Estimate n_γ today, n_γ/n_b , decoupling temperature, electron relativity, $a(t)$ in matter era, $T(t)$, age at decoupling.

(a) Nucleosynthesis vs decoupling

Big-Bang nucleosynthesis (BBN). Epoch $T \sim 0.1\text{--}1\text{ MeV}$, $t \sim 1\text{--}300\text{ s}$. Universe radiation-dominated. Weak interactions freeze out at $T_{\text{wfo}} \sim 0.8\text{ MeV}$, fixing $n/p \approx 1/6$ (later $\rightarrow 1/7$ from neutron decay). Once the photon bath cools below the deuteron binding ($\sim 0.07\text{ MeV}$, photodissociation barrier; the actual nuclear chain proceeds at $T \sim 70\text{ keV}$ due to high η^{-1} entropy/baryon ratio — the “deuterium bottleneck”), nuclei build up: $D, {}^3\text{He}, {}^4\text{He}, {}^7\text{Li}$. Primordial mass fractions: $Y_{\text{He}} \approx 0.25$, $D/H \sim 10^{-5}$.

Decoupling (recombination). Epoch $T \sim 0.3\text{ eV}$, $t \sim 3.8 \times 10^5\text{ yr}$. Photons cease Thomson-scattering off free electrons because the electrons combine with protons to form neutral hydrogen: $e^- + p \rightarrow H + \gamma$. The CMB photons free-stream from the surface of last scattering. The temperature is much below the H ionisation energy 13.6 eV because the high photon-to-baryon ratio $\eta^{-1} \sim 10^9$ keeps a tail of ionising photons until $T \ll 13.6\text{ eV}$.

Common features. Both occur in the radiation-dominated era ($H \propto T^2$, $t \propto T^{-2}$); both are out-of-equilibrium freeze-out events governed by the competition between an interaction rate Γ and the Hubble rate H ($\Gamma \sim H$ defines the freeze-out temperature); both leave fossil relics (light elements, CMB).

Differences. BBN fixes the chemical composition of baryons; decoupling fixes the photon mean free path. BBN is much hotter and earlier, governed by strong/weak nuclear physics; decoupling is governed by atomic physics. Photons are tightly coupled in both, but only at decoupling do they free-stream.

(b) CMB spectrum and present temperature

The CMB is a near-perfect blackbody at

$$T_0 = 2.725\text{ K}.$$

Spectral form: Planck distribution

$$u_\nu d\nu = \frac{8\pi h\nu^3/c^3}{e^{h\nu/k_B T} - 1} d\nu.$$

Typical photon energy: $E_\gamma \sim 2.7k_B T$ (mean for blackbody). Compute:

$$k_B T_0 = 1.381 \times 10^{-23} \times 2.725 = 3.76 \times 10^{-23}\text{ J}.$$

$$E_\gamma \approx 2.7 \times 3.76 \times 10^{-23} = 1.02 \times 10^{-22}\text{ J}.$$

Photon number density (energy density / energy per photon):

$$n_\gamma \sim \frac{u_\gamma}{E_\gamma} = \frac{4 \times 10^{-14}}{1.02 \times 10^{-22}}.$$

Divide:

$$n_\gamma \approx 3.9 \times 10^8\text{ m}^{-3} \approx 4 \times 10^8\text{ m}^{-3}.$$

(Exact Planck integration gives $4.1 \times 10^8\text{ m}^{-3}$.)

(c) Photon-to-baryon ratio and decoupling temperature

Critical density: $\rho_c = 3H_0^2/(8\pi G) \approx 9 \times 10^{-27} \text{ kg m}^{-3}$.

Baryon density (using $\Omega_b \approx 0.05$):

$$n_b = \Omega_b \rho_c / m_p = 0.05 \times 9 \times 10^{-27} / 1.67 \times 10^{-27} = 0.27 \text{ m}^{-3}.$$

Photon-to-baryon ratio:

$$\eta^{-1} = n_\gamma / n_b = 4 \times 10^8 / 0.27 \approx \boxed{1.5 \times 10^9}.$$

Decoupling temperature. Saha-style estimate: recombination occurs when the fraction of photons in the high-energy tail with $E > E_{\text{ion}} = 13.6 \text{ eV}$ becomes of order $\eta = n_b / n_\gamma \sim 10^{-9}$.

$$\frac{n_\gamma(E > E_{\text{ion}})}{n_\gamma} \sim e^{-E_{\text{ion}}/k_B T_{\text{dec}}} \sim \eta.$$

Solve:

$$E_{\text{ion}}/(k_B T_{\text{dec}}) \sim \ln(1/\eta) = \ln(1.5 \times 10^9) = 21.$$

$$k_B T_{\text{dec}} \sim 13.6/21 = 0.65 \text{ eV}.$$

$$T_{\text{dec}} \sim 0.65/k_B [\text{eV/K}] = 0.65/8.617 \times 10^{-5} \approx \boxed{7.5 \times 10^3 \text{ K}}.$$

The full Saha calculation gives $T_{\text{dec}} \approx 3000 \text{ K}$; the order-of-magnitude estimate above is high because we used a crude Boltzmann tail and neglected the ionisation-equilibrium prefactor (Saha) and that recombination is more efficient than the simple tail estimate suggests. *Approximations:* (i) Boltzmann tail in place of full Saha; (ii) neglect of two-photon decay channel; (iii) sharp instantaneous recombination.

Are the electrons relativistic? Compare $k_B T_{\text{dec}}$ (or $\sim 0.3 \text{ eV}$ for the accurate value) with $m_e c^2 = 511 \text{ keV}$:

$$k_B T_{\text{dec}} / m_e c^2 \sim 1 \text{ eV} / 511 \times 10^3 \text{ eV} \sim 2 \times 10^{-6} \ll 1.$$

Electrons strongly non-relativistic at decoupling.

(d) Matter era: $a(t)$ and $T(t)$; age at decoupling

Friedmann (flat, matter-dominated, $\rho \propto a^{-3}$):

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \rho = \frac{8\pi G}{3} \rho_0 a^{-3}.$$

Take square root:

$$\dot{a} = C a^{-1/2}, \quad C \equiv \sqrt{8\pi G \rho_0 / 3}.$$

Separate variables:

$$a^{1/2} da = C dt.$$

Integrate:

$$\frac{2}{3} a^{3/2} = Ct.$$

$$\boxed{a(t) \propto t^{2/3} \text{ (EdS, matter era).}}$$

Temperature. Use (??): $T \propto 1/a$ (CMB redshift), so

$$T(t) \propto t^{-2/3}.$$

Age scaling: $t \propto T^{-3/2}$.

Age at decoupling. Use today's age $t_0 \sim 1.4 \times 10^{10}$ yr and $T_0 = 2.725$ K, $T_{\text{dec}} \approx 3000$ K (the textbook value):

$$t_{\text{dec}} = t_0 \left(\frac{T_0}{T_{\text{dec}}} \right)^{3/2} = 1.4 \times 10^{10} \times \left(\frac{2.725}{3000} \right)^{3/2}.$$

Ratio:

$$2.725/3000 = 9.08 \times 10^{-4}.$$

Power 3/2:

$$(9.08 \times 10^{-4})^{3/2} = (9.08 \times 10^{-4}) \times \sqrt{9.08 \times 10^{-4}} = 9.08 \times 10^{-4} \times 3.01 \times 10^{-2} = 2.74 \times 10^{-5}.$$

Multiply:

$$t_{\text{dec}} = 1.4 \times 10^{10} \times 2.74 \times 10^{-5} = 3.8 \times 10^5 \text{ yr.}$$

$t_{\text{dec}} \sim 4 \times 10^5 \text{ yr.}$

(Matter–radiation equality occurs earlier, at $z_{\text{eq}} \sim 3400$, so the strict $t \propto T^{-3/2}$ scaling is good for $z \lesssim 1100$, and the estimate is reliable.)